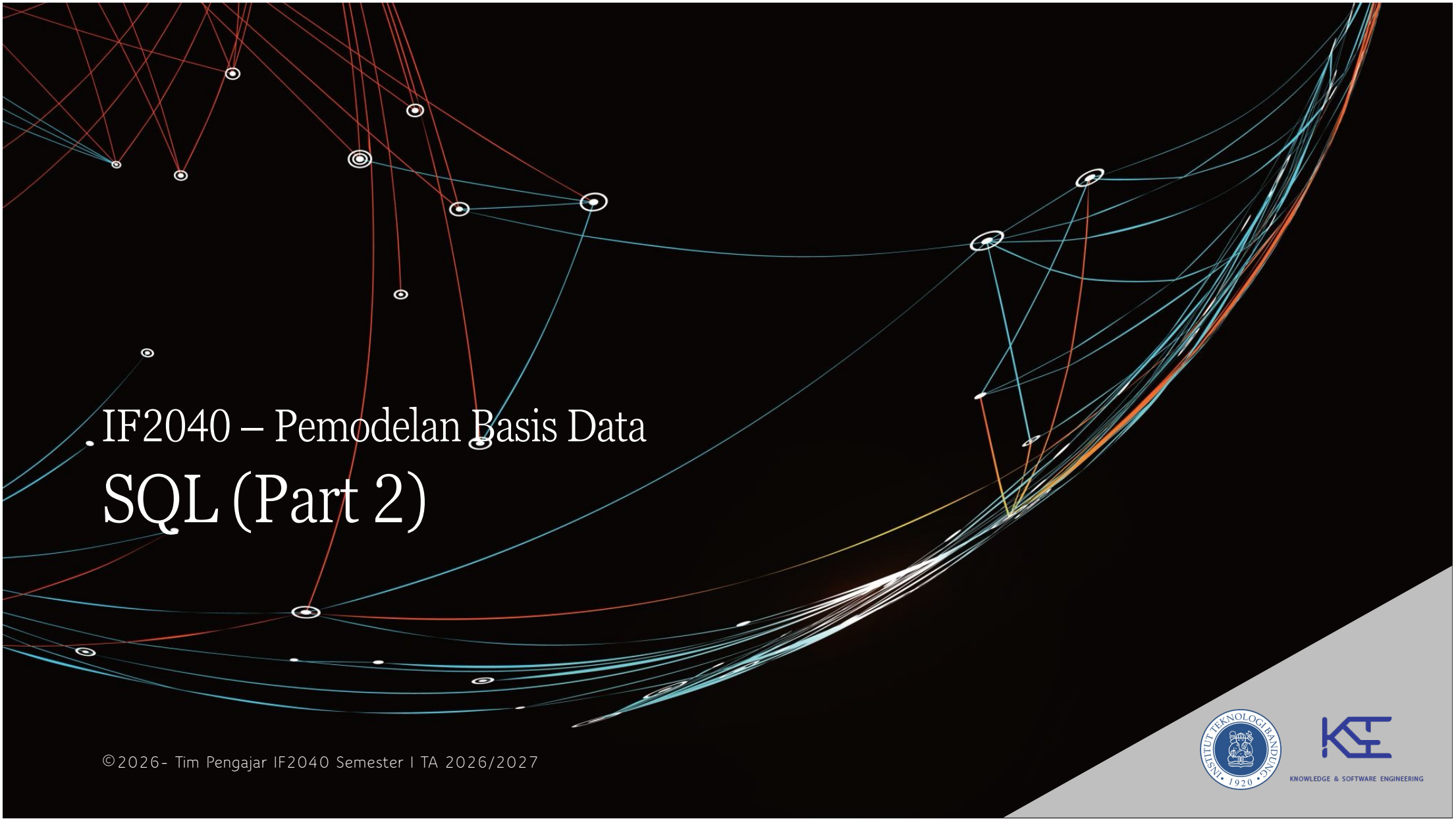


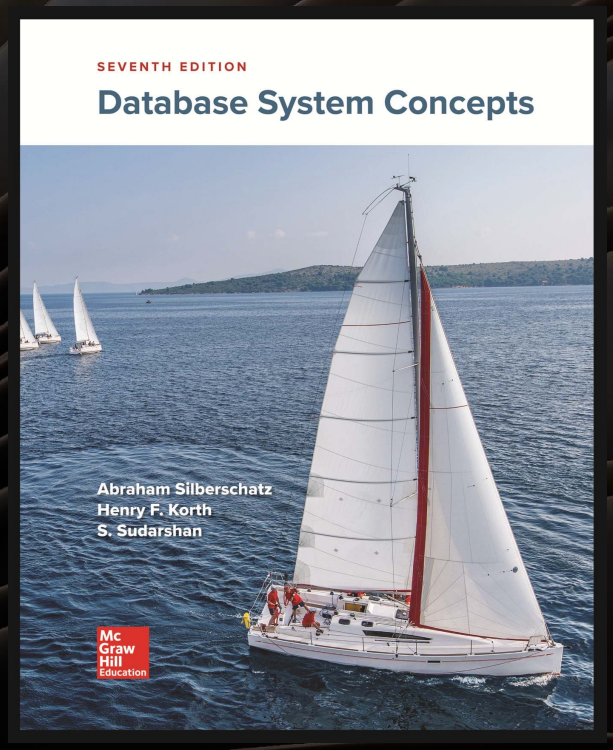
IF2040 – Pemodelan Basis Data

SQL (Part 2)

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Sumber

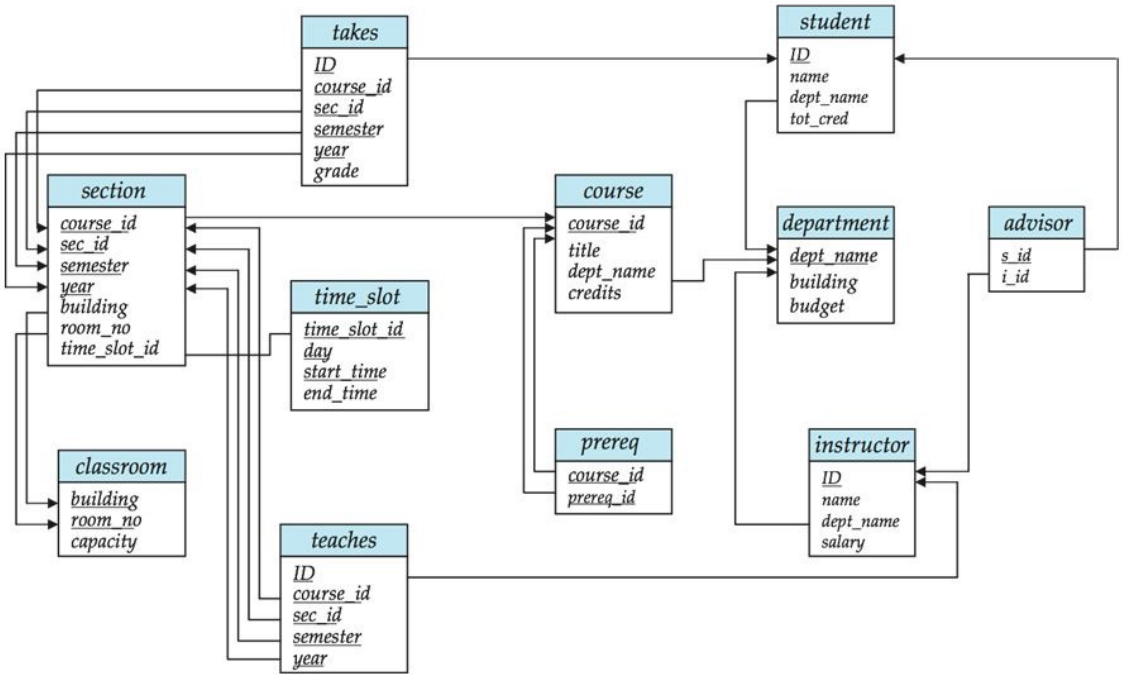
Silberschatz, Korth, Sudarshan: "Database System Concepts", 7th Edition

- Chapter 3 : Introduction to SQL

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Schema Diagram



Aggregate Functions

These functions operate on the multiset of values of a column of a relation, and return a value

avg: average value
min: minimum value
max: maximum value
sum: sum of values
count: number of values





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Aggregate Functions Examples

Find the average salary of instructors in the Computer Science department

- **select avg** (*salary*)
  from *instructor*
  where *dept_name*= 'Comp. Sci.';

Find the total number of instructors who teach a course in the Spring 2018 semester

- **select count** (**distinct** *ID*)
  from *teaches*
  where *semester* = 'Spring' **and** *year* = 2018;

Find the number of tuples in the *course* relation

- **select count** (*)
  from *course*;

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Aggregate Functions – Group By

Find the average salary of instructors in each department

- **select** *dept_name*, **avg** (*salary*) **as** *avg_salary*
  from *instructor*
  group by *dept_name*;

ID	*name*	*dept_name*	*salary*
76766	Crick	Biology	72000
45565	Katz	Comp. Sci.	75000
10101	Srinivasan	Comp. Sci.	65000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
22222	Einstein	Physics	95000

dept_name	*avg_salary*
Biology	72000
Comp. Sci.	77333
Elec. Eng.	80000
Finance	85000
History	61000
Music	40000
Physics	91000

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Aggregation (Cont.)

Attributes in **select** clause outside of aggregate functions must appear in **group by** list

- /* erroneous query */
  select *dept_name*, *ID*, **avg** (*salary*)
  from *instructor*
  group by *dept_name*;

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Aggregate Functions – Having Clause

Find the names and average salaries of all departments whose average salary is greater than 42000

```
select dept_name, avg (salary) as avg_salary
from instructor
group by dept_name
having avg (salary) > 42000;
```

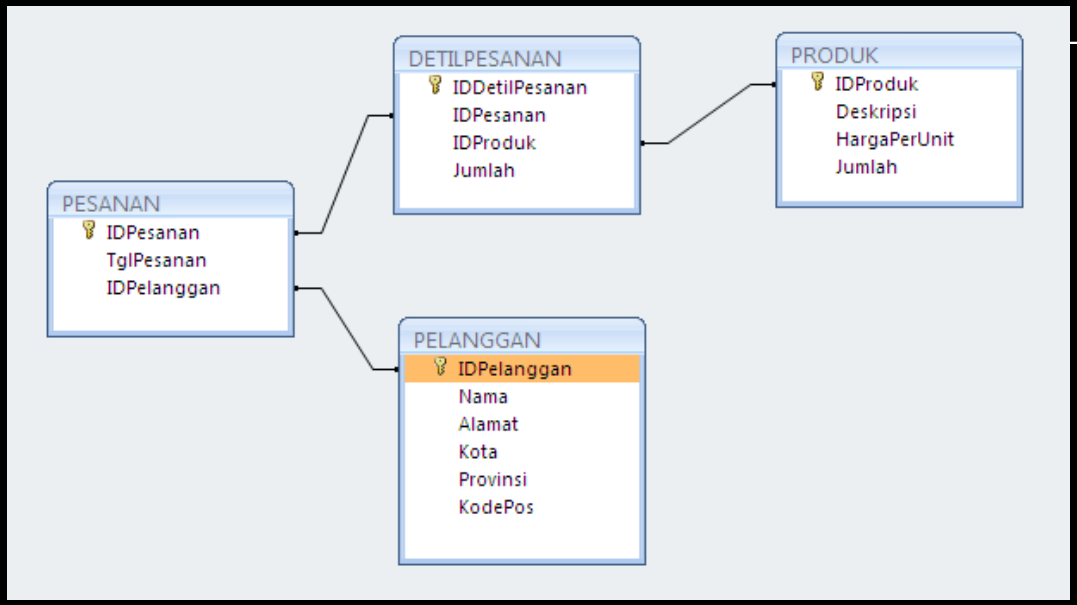
Note: predicates in the **having** clause are applied after the formation of groups whereas predicates in the **where** clause are applied before forming groups





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Latihan-2

1. Tampilkan harga produk tertinggi dan terendah yang pernah dipesan.
2. Tampilkan data nama dan alamat pelanggan yang berasal dari kota "Bandung" serta banyaknya pesanan yang telah dibuat oleh pelanggan tersebut.
3. Tampilkan daftar IDPesanan dan banyaknya item yang dipesan untuk masing-masing pesanan, untuk pesanan dengan banyaknya item lebih dari 10.

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Nested Subqueries

SQL provides a mechanism for the nesting of subqueries. A **subquery** is a **select-from-where** expression that is nested within another query.

The nesting can be done in the following SQL query

select A_1, A_2, ..., A_n
from r_1, r_2, ..., r_m
where P

as follows:

- **From clause:** r_i can be replaced by any valid subquery
- **Where clause:** P can be replaced with an expression of the form:

  B <operation> (subquery)

  B is an attribute and <operation> to be defined later.
- **Select clause:**

  A_i can be replaced be a subquery that generates a single value.

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Set Membership

Find courses offered in Fall 2017 and in Spring 2018

```
select distinct course_id
from section
where semester = 'Fall' and year= 2017 and
        course_id in (select course_id
                      from section
                      where semester = 'Spring' and year= 2018);
```

Find courses offered in Fall 2017 but not in Spring 2018

```
select distinct course_id
from section
where semester = 'Fall' and year= 2017 and
        course_id  not in (select course_id
                           from section
                           where semester = 'Spring' and year= 2018);
```





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Set Membership (Cont.)

Name all instructors whose name is neither "Mozart" nor Einstein"

```
select distinct name
from instructor
where  name not in ('Mozart', 'Einstein')
```

Find the total number of (distinct) students who have taken course sections taught by the instructor with *ID* 10101

```
select count (distinct ID)
from takes
where (course_id, sec_id, semester, year) in
                    (select course_id, sec_id, semester, year
                     from teaches
                     where teaches.ID= 10101);
```

Note: Above query can be written in a much simpler manner.
The formulation above is simply to illustrate SQL features

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Set Comparison – "some" Clause

Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

```
select distinct T.name
from instructor as T, instructor as S
where T.salary > S.salary and S.dept name = 'Biology';
```

Same query using > **some** clause

```
select name
from instructor
where salary > some (select salary
                     from instructor
                     where dept name = 'Biology');
```





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Definition of "some" Clause

F <comp> **some** $r \Leftrightarrow \exists\, t \in r$ such that (F <comp> t)
Where <comp> can be: $<,\ \leq,\ >,\ =,\ \neq$

(5 < **some** | 0 / 5 / 6 |) = true (read: 5 < some tuple in the relation)

(5 < **some** | 0 / 5 |) = false

(5 = **some** | 0 / 5 |) = true

(5 ≠ **some** | 0 / 5 |) = true (since 0 ≠ 5)

(= **some**) ≡ **in**
However, (≠ **some**) ≡ **not in**

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Set Comparison – "all" Clause

Find the names of all instructors whose salary is greater than the salary of all instructors in the Biology department.

```
select name
from instructor
where salary > all (select salary
                    from instructor
                    where dept name = 'Biology');
```


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Definition of "all" Clause

F <comp> all $r \Leftrightarrow \forall\ t \in r$ (F <comp> t)

(5 < **all** | 0, 5, 6 |) = false

(5 < **all** | 6, 10 |) = true

(5 = **all** | 4, 5 |) = false

(5 ≠ **all** | 4, 6 |) = true (since 5 ≠ 4 and 5 ≠ 6)

(≠ **all**) ≡ **not in**
However, (= **all**) ≠ **in**

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Test for Empty Relations

The **exists** construct returns the value **true** if the argument subquery is nonempty.

exists $r \Leftrightarrow r \neq \emptyset$

not exists $r \Leftrightarrow r = \emptyset$

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Use of "exists" Clause

Yet another way of specifying the query "Find all courses taught in both the Fall 2017 semester and in the Spring 2018 semester"

```
select course_id
from section as S
where semester = 'Fall' and year = 2017 and
        exists  (select *
                    from section as T
                    where semester = 'Spring' and year= 2018
                            and S.course_id = T.course_id);
```

Correlation name – variable S in the outer query

Correlated subquery – the inner query

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Use of "not exists" Clause

Find all students who have taken all courses offered in the Biology department.

```
select distinct S.ID, S.name
from student as S
where not exists ( (select course_id
                    from course
                    where dept_name = 'Biology')
                   except
                    (select T.course_id
                     from takes as T
                     where S.ID = T.ID));
```

- First nested query lists all courses offered in Biology
- Second nested query lists all courses a particular student took

Note that $X - Y = \emptyset \quad \Leftrightarrow \quad X \subseteq Y$

Note: Cannot write this query using = all and its variants

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Test for Absence of Duplicate Tuples

The **unique** construct tests whether a subquery has any duplicate tuples in its result.

The **unique** construct evaluates to "true" if a given subquery contains no duplicates .

Find all courses that were offered at most once in 2017

```
select T.course_id
from course as T
where unique ( select R.course_id
                    from section as R
                    where T.course_id= R.course_id
                          and R.year = 2017);
```





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Subqueries in the From Clause

SQL allows a subquery expression to be used in the **from** clause

Find the average instructors' salaries of those departments where the average salary is greater than $42,000."

```
select dept_name, avg_salary
from ( select dept_name, avg (salary) as avg_salary
          from instructor
          group by dept_name)
where avg_salary > 42000;
```

Note that we do not need to use the **having** clause

Another way to write above query

```
select dept_name, avg_salary
from ( select dept_name, avg (salary)
          from instructor
          group by dept_name)
           as dept_avg (dept_name, avg_salary)
 where avg_salary > 42000;
```


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With Clause

The **with** clause provides a way of defining a temporary relation whose definition is available only to the query in which the **with** clause occurs.

Find all departments with the maximum budget

```
with max_budget (value) as
        (select max(budget)
         from department)
select department.name
from department, max_budget
where department.budget = max_budget.value;
```

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Complex Queries using With Clause

Find all departments where the total salary is greater than the average of the total salary at all departments

```
with dept _total (dept_name, value) as
        (select dept_name, sum(salary)
         from instructor
         group by dept_name),
dept_total_avg(value) as
       (select avg(value)
       from dept_total)
select dept_name
from dept_total, dept_total_avg
where dept_total.value > dept_total_avg.value;
```





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Scalar Subquery

Scalar subquery is one which is used where a single value is expected

List all departments along with the number of instructors in each department

```
select dept_name,
        ( select count(*)
          from instructor
          where department.dept_name = instructor.dept_name)
        as num_instructors
from department;
```

Runtime error if subquery returns more than one result tuple





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